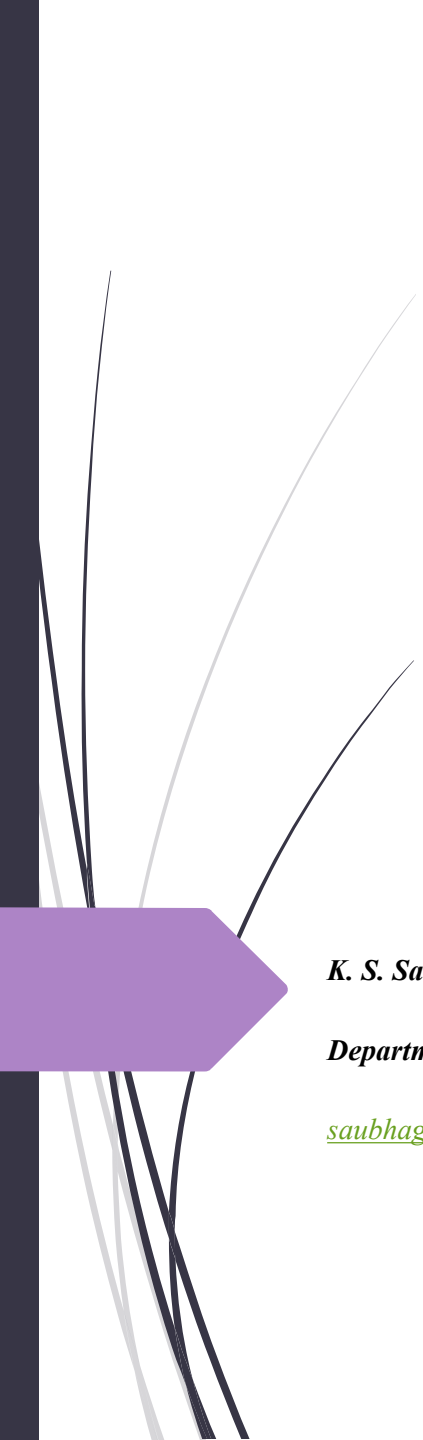




# ST 3007

# Operational Research

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# Transportation Model

## Chapter 4

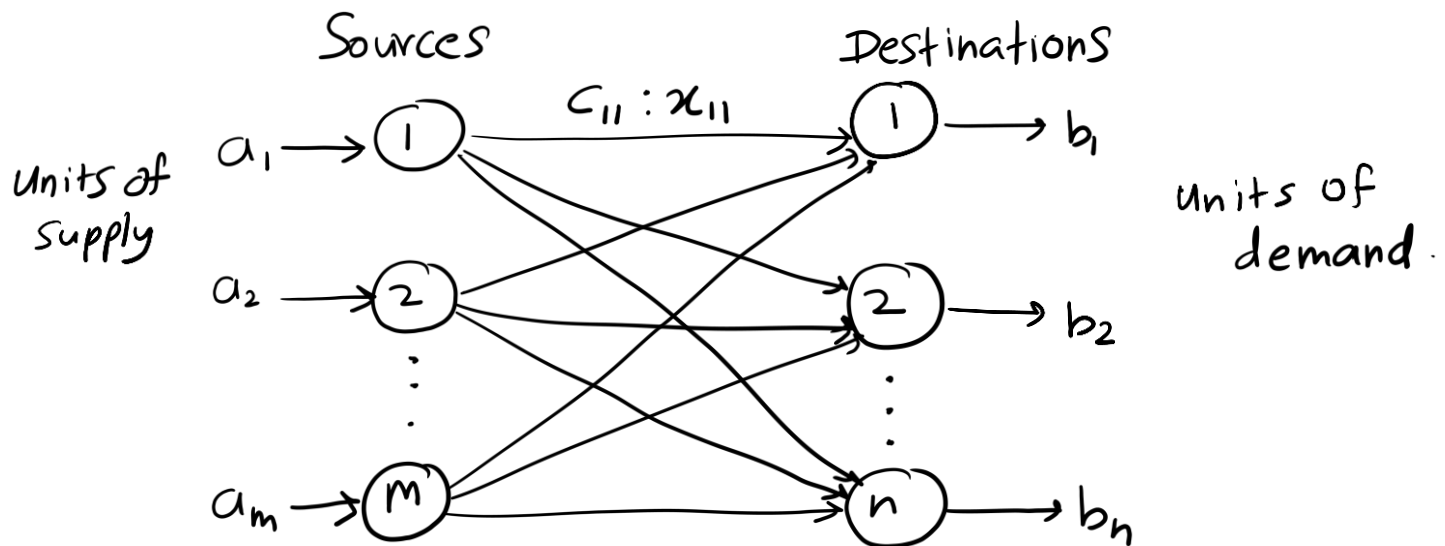
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# Chapter Summary

- **Introduction to Transportation Model**
- **Balancing the transportation model**
- **Transportation algorithm**
- **Determination of the Starting Solution**
- **Iterative Computations of the Transportation Algorithm**
- **Transportation Algorithm: Optimization**

# Introduction

- ▶ Transportation model is a special case of LP model where goods are transported from a set of sources to a set of destinations.
- ▶ The problem is represented by the network in below figure.



# Terms of the Transportation Model

- There are  $m$  sources and  $n$  destinations, each represented by a node.
- The arcs represent the routes linking the sources and the destinations.
- Arc  $(i, j)$  joining source  $i$  to destination  $j$  carries two pieces of information:
  - The transportation cost per unit -  $c_{ij}$
  - The amount shipped -  $x_{ij}$
- The amount of supply at source  $i$  is  $a_i$ , and the amount of demand at destination  $j$  is  $b_j$ .
- The objective of the model is to minimize the total transportation cost while satisfying all the supply and demand restrictions.

## Example: Transportation Model

- MG Auto has three plants in Los Angeles, Detroit, and New Orleans and two major distribution centers in Denver and Miami. The capacities of the three plants are 1000, 1500, and 1200 cars, and the demands at the two distribution centers are 2300 and 1400 cars. The mileage chart between the plants and the distribution centers is given below. Transportation cost is 8 cents per mile per car. Derive the transportation tableau and model the problem as a LP problem.

| Mileage Chart (in miles) |        |       |
|--------------------------|--------|-------|
|                          | Denver | Miami |
| Los Angeles              | 1000   | 2690  |
| Detroit                  | 1250   | 1350  |
| New Orleans              | 1275   | 850   |

# Example: Transportation Model

- The special structure of the transportation problem allows a compact representation of the problem using the transportation tableau format below.

Transportation Model for MG Auto

|             | Denver          | Miami           | Supply |
|-------------|-----------------|-----------------|--------|
| Los Angeles | 80<br>$x_{11}$  | 215<br>$x_{12}$ | 1000   |
| Detroit     | 100<br>$x_{21}$ | 108<br>$x_{22}$ | 1500   |
| New Orleans | 102<br>$x_{31}$ | 68<br>$x_{32}$  | 1200   |
| Demand      | 2300            | 1400            |        |

$x_{ij}$  - Amount of cars shipped from plant  $i$  to distribution center  $j$ .

# Example: Transportation Model

## The LP model of the problem:

- ▶ The optimal solution ships 1000 cars from Los Angeles to Denver, 1300 from Detroit to Denver, 200 from Detroit to Miami, and 1200 from New Orleans to Miami.
- ▶ The associated minimum transportation cost is computed as;

$$1000 * \$80 + 1300 * \$100 + 200 * \$108 + 1200 * \$68 = \$313,200$$

# Balancing the transportation model

- ▶ The transportation tableau representation assumes that **model is balanced**, meaning that the **total demand equals to the total supply** (which happened to be true in the above MG model).
- ▶ If the model is unbalanced, a dummy source or a dummy destination must be added to restore the balance.

## Example:

- ▶ In the previous situation, suppose that the Detroit plant capacity is 1300 cars (instead of 1500).
- ▶ Now the total supply (=3500 cars) is less than the total demand (=3700 cars), meaning that part of the demand at Denver and Miami will not be satisfied.
- ▶ Because the demand exceeds the supply, a dummy plant (source) with a capacity of 200 cars ( $= 3700 - 3500$ ) is added to balance the model.

# Balancing the transportation model

- When adding dummy plant, we set the unit transportation cost from the dummy plant to the two destinations as zero because the plant does not exist.
- Below table gives the balanced model together with its optimum solution.

|             | Denver             | Miami             | Supply |
|-------------|--------------------|-------------------|--------|
| Los Angeles | 80<br><b>1000</b>  | 215               | 1000   |
| Detroit     | 100<br><b>1300</b> | 108               | 1300   |
| New Orleans | 102                | 68<br><b>1200</b> | 1200   |
| Dummy Plant | 0                  | 0<br><b>200</b>   | 200    |
| Demand      | 2300               | 1400              |        |

- We can make sure that a specific destination does not experience shortage by assigning a very high unit transportation cost from the dummy source to that destination.

# Balancing the transportation model

- Now assume demand at Denver is 1900 cars only (supply exceeds the demand).
- Now we add a dummy distribution center to “receive” the surplus supply. Below table gives the new model and its optimal solution to above situation.

|             | Denver            | Miami             | Dummy           | Supply |
|-------------|-------------------|-------------------|-----------------|--------|
| Los Angeles | 80<br><b>1000</b> | 215               | 0               | 1000   |
| Detroit     | 100<br><b>900</b> | 108<br><b>200</b> | 0<br><b>400</b> | 1500   |
| New Orleans | 102               | 68<br><b>1200</b> | 0               | 1200   |
| Demand      | 1900              | 1400              | 400             |        |

- Again, the unit transportation cost to the dummy distribution center is zero. When required a factory to “ship out” completely, a high unit transportation cost is assigned from the designated factory to the dummy destination.

# Transportation algorithm - Introduction

- Today, powerful computer codes can solve transportation models of any size as a regular LP. But there is more to the transportation model than the hand computations.
  1. First, its historical significance in the evolution of OR techniques is important and must be preserved.
  2. Second, the special transportation tableau format can facilitate modeling a number of situations that do not deal directly with transporting goods.
  3. Third, the algorithmic hand computations boast such (almost intuitive) simplicity that a beginner can get a “feel” of what optimization is about.
  4. The transportation algorithm does provide insight into the use of the theoretical primal-dual relationships to achieve a practical end result—that of developing simple rules for hand computations.

# Transportation Algorithm - Steps

- The basic steps of the transportation algorithm are exactly those of the simplex method.
- However, instead of using the regular simplex tableau, we take advantage of the special structure of the transportation model to carry out the algorithmic computations more conveniently.

## Transportation Algorithm:

- **Step 1** - Determine a starting basic feasible solution and go to step 2.
- **Step 2** - Use the optimality condition of the simplex method to determine the entering variable from among all the non-basic variables. If the optimality condition is satisfied, stop. Otherwise, go to step 3.
- **Step 3** - Use the feasibility condition of the simplex method to determine the leaving variable from among all the current basic variables and find the new basic solution. Return to step 2.

# Example: Transportation Algorithm

- SunRay Transport Company ships truckloads of grain from three silos to four mills. The supply (in truckloads) and the demand (also in truckloads) together with the unit transportation costs per truckload on the different routes are give below. The unit transportation costs,  $c_{ij}$  is shown in the northeast corner of each box. The model seeks the minimum cost shipping schedule between the silos and the mills.

|        |   | Mil            |                |                |                | Supply |
|--------|---|----------------|----------------|----------------|----------------|--------|
|        |   | 1              | 2              | 3              | 4              |        |
| Silo   | 1 | 10<br>$x_{11}$ | 2<br>$x_{12}$  | 20<br>$x_{13}$ | 11<br>$x_{14}$ | 15     |
|        | 2 | 12<br>$x_{21}$ | 7<br>$x_{22}$  | 9<br>$x_{23}$  | 20<br>$x_{24}$ | 25     |
|        | 3 | 4<br>$x_{31}$  | 14<br>$x_{32}$ | 16<br>$x_{33}$ | 18<br>$x_{34}$ | 10     |
| Demand |   | 5              | 15             | 15             | 15             |        |

# Determination of the Starting Solution

- ▶ A general transportation model with  $m$  sources and  $n$  destinations has  $m + n$  constraint equations, one for each source and each destination.
- ▶ However, because the transportation model is always balanced, one of the equations is redundant, reducing the model to  $m + n - 1$  independent equations and  $m + n - 1$  basic variables.
- ▶ The special structure of the transportation problem allows securing a non-artificial starting basic solution using one of three methods:
  1. Northwest-corner method
  2. Least-cost method
  3. Vogel approximation method
- ▶ In general, the Vogel heuristic is best, and the northwest-corner method is the worst.
- ▶ The trade-off is that the northwest-corner method involves the least number of computations.

# 1. Northwest-corner Method

- The method starts at the northwest-corner cell (route) of the tableau (variable  $x_{11}$  ).

## Northwest-corner method steps:

- **Step 1** - Allocate as much as possible to the selected cell and adjust the associated amounts of supply and demand by subtracting the allocated amount.
- **Step 2** - Cross out the row or column with zero supply or demand to indicate that no further assignments can be made in that row or column. If both a row and a column net to zero simultaneously, cross out one only, and leave a zero supply (demand) in the uncrossed-out row (column).
- **Step 3** - If all the requirements and availabilities fulfilled, stop. Otherwise, move to the cell to the right if a column has just been crossed out or below if a row has been crossed out. Go to step 1.

# Northwest-corner Method: Implementation

- ▶ The application of the procedure to the model of Example 2 gives the starting basic solution as below.

## 2. Least-cost Method

- ▶ The least-cost method finds a better starting solution by targeting the cheapest routes.

### **Least-cost Method Steps:**

- ▶ First, assign as much as possible to the cell with the smallest unit cost (ties are broken arbitrarily).
- ▶ Next, the satisfied row or column is crossed out and the mounts of supply and demand are adjusted accordingly. If both a row and a column are satisfied simultaneously, only one is crossed out, the same as in the northwest-corner method.
- ▶ Next, select the uncrossed-out cell with the smallest unit cost and repeat the process until all the requirements and availabilities fulfilled.

# Least-cost Method - Implementation

- The application of the procedure to the model of Example 2 gives the starting basic solution as below.

### 3. Vogel Approximation Method (VAM)

- VAM is an improved version of the least-cost method that generally, but not always, produces better starting solutions.

#### **Vogel Approximation Method Steps:**

- **Step 1** - For each row (column), determine a penalty measure by subtracting the smallest unit cost in the row (column) from the next smallest unit cost in the same row (column).
- **Step 2** - Identify the row or column with the largest penalty, breaking ties arbitrarily. Allocate as much as possible to the variable with the least unit cost in the selected row or column. Adjust the supply and demand and cross out the satisfied row or column. If a row and a column are satisfied simultaneously, only one of the two is crossed out, and the remaining row (column) is assigned zero supply (demand).

### 3. Vogel Approximation Method (VAM)

➤ **Step 3 -**

- a) If exactly one row or column with zero supply or demand remains uncrossed out, stop.
- b) If one row (column) with positive supply (demand) remains uncrossed out, determine the basic variables in the row (column) by the least-cost method. Stop.
- c) If all the uncrossed-out rows and columns have (remaining) zero supply and demand, determine the zero basic variables by the least-cost method. Stop.
- d) Otherwise, go to step 1.

# Vogel Approximation Method - Implementation

- ▶ The application of the VAM to the model of previous Example can be conducted as below.

# Iterative Computations of the Transportation Algorithm

- After determining the starting solution (using one of the three methods), we use the following algorithm to determine the optimum solution:
  - **Step 1** - Use the simplex optimality condition to determine the entering variable. If the optimality condition is satisfied, stop. Otherwise, go to step 2.
  - **Step 2** - Determine the leaving variable using the simplex feasibility condition. Change the basis and return to step 1.
- The optimality and feasibility conditions do not involve the familiar row operations used in the simplex method. Instead, the special structure of the transportation model allows simpler (hand) computations.

# Transportation Algorithm: Optimization

## Step 1

- ▶ The determination of the entering variable from among the current non-basic variables is done by computing the non-basic coefficients in the z-row, using the method of multipliers.
- ▶ In the method of multipliers, we associate the multipliers  $u_i$  and  $v_j$  with row  $i$  and column  $j$  of the transportation tableau.
- ▶ For each current basic variable  $x_{ij}$ , these multipliers satisfy the following equations:  
$$u_i + v_j = c_{ij} \text{ for each basic } x_{ij}$$
- ▶ We will use the northwest-corner starting solution to continue with the optimization.
- ▶ The starting solution has six basic variables, which leads to six equations in seven unknowns.

# Transportation Algorithm: Optimization

## Step 1

- Initial Solution from Northwest-corner Method.
- We can give the multiplier equations as below:

# Transportation Algorithm: Optimization

## Step 1

- Next, we use  $u_i$  and  $v_j$  to evaluate the non-basic variables by computing  $u_i + v_j - c_{ij}$  for each non-basic  $x_{ij}$
- The results of these evaluations are shown in the following table:

# Transportation Algorithm: Optimization

## Step 1

- ▶ The preceding information is equivalent to computing the z-row of the simplex tableau, as the following summary shows:
- ▶ Because the transportation model minimizes cost, the entering variable is the one having the most positive coefficient in the z-row (in our case,  $x_{31}$  is the entering variable).
- ▶ All the preceding computations are usually done directly on the transportation tableau, meaning that it is not necessary to write the  $(u, v)$ -equations explicitly.

# Transportation Algorithm: Optimization

## Step 1

- ▶ After evaluating the non-basic variables by computing  $u_i + v_j - c_{ij}$  for each non-basic  $x_{ij}$ , we show those values in the boxed southeast corner of each cell as below.

# Transportation Algorithm: Optimization

## Step 2

- Having identified the entering variable ( $x_{31}$ ), we need to determine the leaving variable.
- The selection of  $x_{31}$  as the entering variable means shipping through this route reduces the total shipping cost. Hence, we need to maximize  $x_{31}$ .
- Note that if route (3, 1) ships  $\theta$  units (i.e.,  $x_{31} = \theta$ ), then the maximum value of  $\theta$  is determined based on two conditions:
  1. Supply limits and demand requirements remain satisfied.
  2. Shipments through all routes remain nonnegative.
- These two conditions determine the maximum value of  $\theta$  and the leaving variable in the following manner.
  - First, construct a closed loop that starts and ends at the entering variable cell.
  - The loop consists of connected horizontal and vertical segments only (no diagonals) whose corner elements (excluding the entering variable cell) must coincide with a current basic variable.

# Transportation Algorithm: Optimization

## Step 2

- Table shows the loop for our entering variable  $x_{31}$ . Exactly one loop exists for a given entering variable.
- Next, we assign the amount  $\theta$  to the entering variable. For the supply and demand limits to remain satisfied, we must alternate between subtracting and adding the amount  $\theta$  at the successive corners of the loop.

# Transportation Algorithm: Optimization

## Step 2

- Table below shows the closed loop with  $\theta$  (it is immaterial if the loop is traced in a clockwise or counterclockwise direction).

- For  $\theta \geq 0$ , the new values remain non-negative if;

$$x_{11} = 5 - \theta \geq 0$$

$$x_{22} = 5 - \theta \geq 0$$

$$x_{34} = 10 - \theta \geq 0$$

# Transportation Algorithm: Optimization

## Step 2

- ▶ The corresponding maximum value of  $\theta$  is 5, which occurs when both  $x_{11}$  and  $x_{22}$  reach zero level. Therefore, we can take either  $x_{11}$  or  $x_{22}$  as the leaving variable.
- ▶ Normally we break the ties by selecting the variable with the higher unit cost. Hence, we choose  $x_{11}$  as the leaving variable.
- ▶ Now adjust the values of the basic variables at the corners of the closed loop to accommodate setting  $x_{31} = 5$  as below.

# Transportation Algorithm: Optimization

## Step 2

- Given the new basic solution, we repeat the computation of the multipliers  $u$  and  $v$  as before and identify the entering variable, which is  $x_{14}$ .
- Again, identify the closed loop and determine the leaving variable.

# Transportation Algorithm: Optimization

## Step 2

- ▶ We repeat these two steps until optimal solution is derived (new values of  $u_i + v_j - c_{ij}$  are negative for all non-basic  $x_{ij}$ ).
- ▶ **Optimal Solution:**  $x_{12} = 5; x_{14} = 10; x_{22} = 10; x_{23} = 15; x_{31} = 5; x_{34} = 5$ ; Optimal cost = 435.

# Thank You!

**Next Chapter - Degeneracy in Transportation Model  
and Transshipment Model**